Low-Earning Workers and Social Security: Evidence from Costa Rica's Contributions Reform

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Abstract

Social security contributions raise the cost of formal employment and are considered a key determinant of informality among low-wage workers in developing economies, yet evidence on whether reducing them increases registration is mixed. In Costa Rica, a minimum contribution floor makes registration particularly costly for these workers and their employers. In 2023, a reform lowered this floor, reducing contributions for both employers and employees, with a gradual rollout by age. Using administrative and survey data and a regression discontinuity design that exploits the phased implementation, we evaluate the reform's effects on registration and other labour market outcomes six months after it came into force. We find no significant effects on registration or on earnings reported to the social security administration. However, using constructed measures of net and gross earnings, we find higher take-home pay for workers and lower total labour costs for firms, with cost savings shared in line with the statutory changes. These short-term results suggest that lowering contribution costs alone may not be enough to increase registration, and that complementary policies are needed to address remaining barriers to formality.

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1. Introduction

Payroll-financed social security systems remain a key mechanism through which societies provide insurance against risks such as illness and old age. For salaried workers, access to benefits like pensions and health coverage often depends on being registered with the national social insurance system and making regular, often substantial, social security contributions (SSCs) based on declared earnings, as mandated by national legislation. However, globally only six out of ten employees are registered with the social security system (ILO 2023). In Latin America, less than half of workers are registered (ILO 2018), and registration tends to be especially low for those at the bottom of the earnings distribution (Levy and Cruces 2021; Lopez-Acevedo et al. 2023).[1] One of the most commonly cited barriers to registration is the cost of payroll taxes, including SSCs, which can be especially burdensome for low earners (Perry 2007). This can undermine revenue collection through SSC and restrict the populations' access to insurance benefits if these are tied to SSC compliance.

This paper aims to evaluate the impact of reducing SSCs for low-earning employees on their registration rates, earnings and other labour market outcomes. To do this, we leverage a recent reform in Costa Rica that lowered a minimum contribution floor for SSCs, considerably reducing contribution costs for both low-earning employees and their employers, with the goal of increasing registration. The reform was implemented gradually through an age-based rollout.

Costa Rica requires all employees to be registered with the social security administration, but close to 20% for salaried workers remains unregistered, and around half of the total workforce (INEC 2023). As many other Latin American and middle-income countries, Costa Rica applies a minimum contribution floor for SSCs. [2] This floor, known as the Minimum Contributory Base (MCB), is set close to the monthly minimum wage (MMW), the wage earned by a full-time employee receiving the legal minimum hourly rate. For employees whose earnings fall below the MCB, both employer and employee contributions must be calculated as if the employee earned the full MCB. In practice, this leads to higher effective contribution rates for low-earning registered employees

[1] Employees whom are not registered with social security through their jobs are also referred to in the literature as informal employees.

[2] Examples include Egypt, Morocco and Tunisia in Africa (Lopez-Acevedo et al. 2023), and Argentina, Mexico and Colombia in Latin America (Levy and Cruces 2021).

and their employers. This elevated cost is widely recognised as a key barrier to the registration of low-earning employees. It also carries important implications for equity, efficiency, and fiscal revenue.

Responding to these challenges, Costa Rica introduced a reform in 2023 to reduce the MCB in a phased way—starting with workers under age 35, then extending to those under 50, and finally to all workers. The aim was to lower the costs associated with registering low-earning employees and encourage greater participation in the system. Taking advantage of the rollout design of this recent reform and rich administrative employer-employee data and household labour surveys, we aim to quantify the impact of lowering MCB on registration, earnings and other labour outcomes using regression discontinuity and difference-in-difference designs. At this stage, our analysis treats SSCs in Costa Rica as a pure payroll tax for simplicity, rather than as contributions tied to specific benefits. This assumption would be appropriate if the actual or perceived links between contributions and benefits are weak. Further exploration of the system is needed to define the most appropriate approach (see Section 2 for more details).

Our preliminary findings based on administrative data suggest the absence of short-term effects of the reform on registration and earnings reported to the social security among these workers. Taking the preliminary findings at face value, they are consistent with the idea that structural features — such as workers' alternative access to healthcare or low salience of pension benefits in the short-term, the slow adjustment of employer practices, or other binding constraints — may undermine the impact of this type of cost-based reforms, at least in the short-run. However, when we look at the calculated net and gross earnings, we observe higher take-home pay for employees and lower total labour costs for employers. This suggests the the economic incidence of the reform followed its statutory incidence. Further analysis using survey data that includes both registered and unregistered employees, along with a deeper understanding of employment patterns among employees at the lower end of the earnings distribution over time, will be essential to fully assess the impact of the reform. We plan to extend our analysis in this direction once access to panel labour force survey data is granted.

This paper aims to make several contributions. First, it adds to the literature on the impact of payroll tax on labour market outcomes. Existing evidence on the employment and incidence effects of payroll taxes cuts remains mixed. Some studies find

no significant employment effects with the benefits of payroll tax changes passed on to employees in the form of higher earnings (Gruber 1997; Gruber and Krueger 1991; Gruber 1994; Cruces et al. 2010). Others report the opposite: increased employment but no significant change in employees' earnings (Saez et al. 2019; Kugler et al. 2017a)). A third group finds partial pass-through of tax changes to workers (Hamermesh 1979; Holmlund 1983; Kugler and Kugler 2009a). Looking more specifically at the effects of social security contributions, the effects tend to be significant at the extensive margin of registration, while mixed on the earnings side (Kugler and Kugler 2009b; Kugler et al. 2017b; Bergolo and Cruces 2014). This divergence may be due in part to differences in institutional contexts, the design of systems, and the specific outcomes examined, often determined by the availability of data. While most of this literature focuses on reforms affecting workers across the entire earnings distribution (e.g., Gruber 1997; Hernandez 2024), or those at the upper end (e.g., Saez et al. 2012), our contribution lies in analysing a reform that specifically targets low-earning employees.

Second, and although a substantial literature studies the effects of other minimum thresholds— notably the minimum wage (e.g., Cengiz et al. 2019; Card and Krueger 2000)—to the best of our knowledge, this is the first paper to analyse and causally estimate the effects of reducing a minimum contribution floor for social security. Despite their prevalence across several countries and their recognised distortionary potential and disincentives for registration with social security (Levy and Cruces 2021; Guerrero 2020), minimum contribution floors remain understudied in the literature. Finally, this study also aims to inform the policy debate on how to design inclusive yet fiscally sustainable social security systems, particularly in settings characterised by low registration rates, large shares of low-earning workers, multiple and often distortionary labour regulations, weak enforcement, and tight budgetary constraints. Specifically, it offers insights into whether reducing SSCs can effectively increase registration, and what impact such a change may have on other labour market outcomes.

This rest of this paper is organised as follows. Section 2 describes Costa Rica's social security institutional framework and the details of the 2023 reform. Section 3 introduces the data used in our analysis. Section 4 outlines the conceptual framework that guides our empirical approach. Section 5 explains the empirical strategy employed. Section 6 presents the main findings. Finally, Section 7 concludes and discusses next steps.

2. Institutional Framework

2.1. Social Security System in Costa Rica

Costa Rica has a long-established social security system dating back to 1941, administered by the *Caja Costarricense del Seguro Social* (CCSS). The system is funded through monthly contributions from employers, employees, and self-employed workers, along with additional support from the government. Registration and making regular SSC provide workers access to two benefits: public healthcare, delivered through the *Seguro de Enfermedad y Maternidad* (SEM), and public pensions, provided through the *Seguro de Invalidez, Vejez y Muerte* (IVM). Healthcare benefits are not proportional to contributions —all contributors are entitled to care without a cap—, while pension benefits are linked to contribution amounts.

Even though the law requires all employees to be registered to access social security benefits, unregistered workers may still in some cases access public healthcare, for example, through a household member (such as a spouse) or through state-sponsored insurance for vulnerable groups. However, they do not accumulate rights toward a contributory pension in old age, nor do they benefit from other formal employment-related protections such as the minimum wage enforcement or lump-sum payments in case of unemployment. Furthermore, for registered employees, access to pensions are contingent on a high level of regular contributions that may be too onerous for low earners (300 monthly contributions, equivalent to 25 years). In addition, the long time horizon may make pensions feel disconnected from the immediate needs of younger workers. These features may lower incentives to contribute at the individual level.

2.2. Employer and Employee Contribution Scheme

By law, all employees —defined as salaried workers who have an employment relationship with an employer involving a set schedule, specific responsibilities, and remuneration, and the group of workers we focus on in this paper— are required to be registered and contribute to the social security system, together with their employer, regardless of their earnings level.[3] Hence, if an employer is found not to have registered their employees, they may face retroactive contribution charges, fines, and even imprisonment.

[3] Occasional workers, i.e. short-term or one-off employees, are exempt from this rule.

Employer and employees' SSCs are calculated using flat statutory percentage rates of an employee's reported posted earnings. We define posted earnings, in line with the terminology of Saez et al. (2012), as the employee's earnings (reported by the employer) before deductions for their SSCs —that is, the amount shown on the payslip prior to any withholdings— and excluding the employer's contributions.

However, the system also includes a Minimum Contribution Base (MCB) rule, which sets a floor on the earnings base used to calculate contributions. The MCB is defined as a fraction, close to one, of the statutory monthly minimum wage (MMW).[4] As a result, when an employee's posted earnings fall below the MCB, both employer and employee contributions are computed as if the worker earned exactly the MCB. Importantly, only part-time employees (defined as those working fewer than 48 hours per week) are allowed to be registered and report posted earnings below the MCB, at least on paper, since the law only permits this group to declare earnings under the MMW, as long as they comply with the minimum hourly wage.

Table 1 summarises the effective average contribution rates (EACR), and reflects the implications of the application of the MCB prior to the 2023 reform, broken down by employer and employee, and by healthcare and pension components. The schedule includes two separate MCB thresholds: for healthcare contributions, the MCB is set at $\text{MCB}^H = 0.87 \cdot \text{MMW}$, and for pension contributions, at $\text{MCB}^P = 0.93 \cdot \text{MMW}$. In 2022, the value of the MCB^H was 287,360 Costa Rican colones (approximately 477 US dollars), while the MCB^P was 307,013 colones (around 510 US dollars). As shown in the table, employers total statutory contribution rate (healthcare + pensions) was of 14.5%, while for employees 9.5%, and the MCB rule applied to both.

[Table 1 here]

The MCB rule causes workers earning below the thresholds, as well as their employers, to face a fixed cost of registration that translates into a higher EACR compared to those earning above it. Moreover, the lower an employee's posted earnings fall below the MCBs, the higher this effective rate becomes. This is illustrated by the solid red lines in Figure 2, which show the EACR for employees and employers at different levels

[4] Costa Rica sets different MMWs based on the qualifications required for each job. The MCB is calculated using the lowest of these, the MMW for the unskilled worker category.

of posted earnings before the implementation of the MCB reform. These rates applied to all salaried work contributions.

Although the MCB was introduced to help ensure a minimum income for workers —presumably under the assumption that most employees work full-time hours— it is widely considered a key barrier to the registration of low earners. This design results in relatively higher registration costs for workers earning below the MCB, creating a potential disincentive to register. This, along with other factors such as a likely binding MMW, may partly explain the pattern observed in registration rates by posted earnings level. Panel A of Figure 1 shows that employees with monthly posted earnings below MCB^H have significantly lower registration rates: while more than 10% of employees earning at or above MCB^H are registered, this share falls to just 22% among those earning less. Panel B further shows that employees working fewer than full-time weekly hours face the highest rates of non-registration, exceeding 60%. Together, these patterns suggest that the MCB poses a substantial barrier to registration for low-earning part-time employees.

[Figure 1 here]

2.3. The MCB Reform

In January 2023, Costa Rica implemented a reform to lower the cost of social security registration for employees earning less than the MCBs. The change came given the concerns about how the MCB discouraged registration of low-earning employees, and followed recommendations of international organisations such as the OECD (Guerrero 2020; OECD 2017). The goal of this reform was to ease the burden for registering low-earners and boost their registration. The reform targeted part-time employees, the only group allowed to report earnings below the MCBs. [5]

The main change introduced by the reform was that the MCBs were cut in half [6]. The reform was rolled out gradually: starting in 2023 with employees under age 35 (first wave), expanding to those under age 50 in 2024 (second wave), and finally applying to all

[5] Self-employed workers were excluded from this reform, based on the argument that they are not subject to fixed working hours and that by law they are not required to contribute when their income falls below the MCB (CCSS 2022).

[6] To be able to register below the MCBs, an employee must be declared by their employer to be working less than full-time.

employees in 2025 (third wave). We refer to the cohorts affected by the reform as treated, and to those not yet affected at a given time as untreated. The changes described below were thus implemented progressively by age cohort over the three-year period.

For healthcare contributions, a new reduced MCB equal to 50% of the original was introduced $\text{MCB}^{H}_{new} = 0.5 \cdot \text{MCB}^{H}$. For employees with posted earnings below the old MCB^{H} and above the new MCB^{H}_{new}, contributions are paid on their posted earnings using the statutory rate. For those below the MCB^{H}_{new}, they need to pay as if they were earning the (lower) MCB^{H}_{new}.

For pensions, similar changes were made, with a new threshold of $\text{MCB}^{P}_{new} = 0.5 \cdot \text{MCB}^{P}$, but the system for those below MCB^{P}_{new} became more proportional. Pension contributions for employees with monthly earnings below MCB^{P}_{new} are based on the closest of four bands (25%, 50%, 75%, or 100% of the new, lower MCB^{P}_{new}). Employees are credited with pension quotas of 0.25, 0.50, 0.75, or 1.0, depending on the percentage on which contributions were made. Lastly, as part of scheduled periodic adjustments unrelated to the MCB reform, statutory pension contribution rates were slightly increased for all employees' contributions in January 2023: the employer rate rose from 5.25% to 5.42%, and the employee rate from 4.00% to 4.17%. These changes were small and affected both treated and untreated cohorts in January 2023. We do not believe they will interfere with our empirical strategy and results, but further checks will be needed.

Panels A and B of Figure 2 show the EACR for both employers and employees before the reform (solid red lines) and in 2023, after the first wave of the reform was implemented (blue dashed lines). At that moment, the reform applied only to contributions of employees younger than 35, which is the variation we explore currently given data availability. The dark blue dashed lines represent those not yet treated by the reform in 2023 (35 and older), while the light blue dashed lines represent the treated group (under 35). The increases in the statutory pension contributions are incorporated here, but given their small size and the scale of the figures, they barely noticeable in the figure. The gap between the dark and light blue dashed lines below the full MCBs reflects the substantial decrease in registration costs both for treated employers and employees.

[Figure 2 here]

2.4. Interactions with Other Taxes

It is also important to consider how other taxes interact with the SSCs, with the MCB rule, and with the system's reform. On the one hand, interactions with the Personal Income Tax (PIT) are not a concern, as the PIT exemption threshold is set at 2.6 times the MCB^P, meaning it does not bind for low-earners. On the other hand, other payroll taxes—such as mandatory private pension contributions (levied on both employers and employees), and additional taxes used to fund social programmes and public institutions (levied only on employers)—raise the overall effective tax burden on all registered salaried employment. Importantly, the MCB rule does not apply to these other taxes. While we recognise the relevance of this broader tax context, we plan to incorporate these interactions more fully in future versions of this study. Table 2 presents the statutory rates for all payroll taxes in Costa Rica.

[Table 2 here]

3. Data

Administrative data. The administrative social security data comes from the matched employer-employee panel dataset *(Sistema Centralizado de Recaudación – SICERE)*, constructed from monthly SSC records maintained by the social security administration (*Caja Costarricense de Seguro Social – CCSS*), and facilitated by the Central Bank of Costa Rica. The complete dataset available spans from January 2006 to December 2024 and covers all registered employers, employees, and self-employed workers. Currently, we have access to data from January 2021 to December 2023, but we expect to expand this to other years soon. Worker-level variables include anonymised personal ID, reported monthly posted earnings, occupation, date of birth, marital status, gender, country of birth, and geographic identifiers of place of residence. On the employer side, the dataset includes anonymised firm identifiers and firm-level geographic information.[7] Each observation in this dataset corresponds to one individual job registration, and workers can have multiple jobs during the same month.

At this stage, we limit the sample to employees with only one registered job, to avoid complications from multiple employment records. We also restrict the sample to

[7] We are currently exploring the possibility of gaining access to additional variables, such as hours worked and firm-level economic activity.

registered employees aged between 25 and 45 as of January 2023. Most of the analysis uses administrative data from June 2023. This gives us six months after the first wave MCB reduction for workers under age 35, which was introduced in January 2023, and six months before the next wave of the reform, which extended the reduction to workers under age 50 in January 2024. However, we also use data from January to June 2023 when looking at new registrations over that period. Finally, we focus the majority of our analysis on the subsample of salaried workers with reported earnings below MCB^P. Table 3 shows the summary statistics for this sample in June 2023.

[Table 3 here]

Survey data. Second, we use survey data from the Labour Force Survey *(Encuesta Continua de Empleo – ECE)*, conducted by the National Institute of Statistics and Census *(Instituto Nacional de Estadística y Censos – INEC)*. This representative quarterly household labour survey has been conducted since the third quarter of 2010 and follows a rotating semi-panel design: each household remains in the survey for four consecutive quarters before being replaced, with roughly 25,000 observations collected each quarter. The survey captures detailed labour market and socioeconomic information at both the household and individual levels, allowing us to follow individuals over one year. Variables include weekly earnings, weekly hours worked, registration status with the social security administration (registered or unregistered), occupation, economic activity, gender, age, geographic identifiers, employer characteristics, and other relevant socioeconomic indicators. An advantage of this dataset is to be able to observe the individuals that are not registered with the social security, which is not possible using the administrative data. Moreover, this data set allows us to see a workers *actual* earnings, not only the one reported reported by their employer. This data spans from the third quarter of 2010 to the fourth quarter of 2024, including data for salaried and self-employed work. While the cross-sectional data is publicly available for download, we are currently awaiting access to the panel version with individual identifiers, which will allow us to track individuals over time. So far, we use this data for descriptive purposes, and have not yet used it to conduct causal analysis.

4. Conceptual Framework

'Explain this in words Link between benefits and contributions?

This section presents a preliminary conceptual framework to guide our empirical analysis of registered labour demand and supply decisions, as well as the incidence effects of the reduction in the MCB. Throughout this section and the rest of the paper, we adopt the terminology of Saez et al. (2012) to refer to different definitions of registered employees' earnings: (i) net earnings (take-home pay, after employee SSC), (ii) posted earnings (the earnings shown on payslips before deductions), and (iii) gross earnings (posted earnings plus employer SSC, reflecting total labour costs[8]).

4.1. Employers' Decisions

[Work-in-progress]

4.2. Employees' Decisions

If the MCB reform changed how much younger workers take home from registered jobs compared to older ones, we would expect to see labour supply responses both along extensive and intensive margins. Here we present a preliminary framework to think about how individuals decide whether to take a job that includes registration with the social security system, and how much to work if they do. Also, we analyse implications of a reform that reduces an MCB on these two outcomes.

We assume individuals can either take registered employment or opt for the outside alternative, which includes unregistered work or relying on government transfers. Individuals have preferences over two dimensions: take-home pay — which captures the value of consumption — and posted earnings — which reflect the hours involved in registered work. An individual's utility function depends on these preferences and their level of ability.[9] Differences in preferences and ability across individuals lead to heterogeneity in the decision to take up registered employment, and how much to work. The decision to register depends on how the utility from registered work compares to that of the outside option, and the shape of the budget set. Those who choose registered employment select their preferred combination of posted earnings (work) and net earnings (take-home pay).

[8] At this stage, we abstract from other payroll taxes.

[9] At this stage, we abstract in this analysis from workers' valuation of the benefits from registered work, such as public healthcare and pension rights, though this will be incorporated in upcoming versions.

Figure 3 illustrates the registration (extensive) and how much to work (intensive) decisions, and the implications of a cut in the MCB. Pre-reform, the budget set corresponds to the solid light blue line. It includes a kink when posted earnings (y) equal the MCB, which arises from the contribution rule: for posted earnings below the MCB, workers pay a fixed amount based on the MCB, $\text{MCB} \cdot \tau^l$, so their net earnings equal $y - \text{MCB} \cdot \tau^l$, where τ^l is the employee statutory contribution rate. In this region, the budget line has a slope of 1. For posted earnings above the MCB, contributions are proportional, so net earnings equal $y \cdot (1 - \tau^l)$, with slope $1 - \tau^l$.

[Table 3 here]

We perform the analysis for three types of individuals (Type 1, Type 2, Type 3) who differ in their utility function. Pre-reform, given their preferences and ability, the Type 1 individual (Panel A in Figure 3) locates at A, at a point tangent to the solid blue line, where they register and have posted earnings above the MCB. Type 2 (Panel B) also registers at point C, with posted earnings just below the MCB. However, the Type 3 (Panel C) individual prefers not to register and chooses the outside option B, as registering at point F would place them on a lower indifference curve.

Post-refrom, where the MCB is halved to MCB_{new}, the kink in the budget set shifts left. For posted earnings below the new MCB, the cost of registering drops, as reflected in the dashed light blue line showing the new budget constraint. For posted earnings above the MCB, the budget line remains the same. The reform does not change the decision of the Type 1 individual, who continues to register and work the same. The Type 2 individual continues to register as well, but given the change in the budget set, they prefer to work fewer hours (stronger income effect) and locate at point D. Furthermore, the Type 3 individual is now indifferent between registering at a new point E or taking the outside option B, as both lie on the same indifference curve. In this case, the reform makes registered work more attractive for low earners who were previously better off remaining unregistered. Thus, theoretically, the reduction of registration costs could lead to adjustments in labour supply both at the intensive margin (working fewer or more hours, depending on their utilities) and at the extensive margin (more workers choosing to register).

At this stage of the paper, based on the outcomes discussed above, we focus on the

registration decision. In future versions, we plan to explore other extensive margin outcomes, such as exiting the labour market or transitioning to self-employment, as well as intensive margin outcomes like hours worked.

4.3. Implications on Earnings

We now examine how the benefits from reducing the MCB may have been shared between treated registered employers and employees. If neither side fully absorbs the cost reduction, we would see both an increase in net earnings (reflecting gains for employees) and a decrease in gross earnings (reflecting savings for employers). Moreover, any change in posted earnings would provide information on how the cost savings were shared. In social security systems, contributions are typically split nominally between employers and employees. However, standard incidence analysis suggests this statutory division does not necessarily reflect the true economic incidence of the cost change.

To think about how the cost reduction from the reform is shared between employers and employees, we follow the logic in Saez et al. (2012). Their theoretical framework considers the case of a payroll tax increase and argues that in a frictionless labour market, if workers just above and below a policy threshold are close substitutes—meaning they have similar characteristics and thus productivity—then employers should not accept differences in total labour costs between them. Consequently, any statutory increase in contributions for one group should be fully passed on to employees (regardless of who is legally responsible for paying the tax), resulting in lower posted and net earnings, while gross earnings remain unchanged. As in our case, the authors focus on comparing workers around the policy threshold, which allows for empirical testing using administrative data. When analysing broader windows with survey data in the future, however, we may need to consider alternative frameworks for understanding how the earnings split evolves after the reform.

In our setting, the reform reduced contributions for the younger, treated group on both the employer and employee sides. This makes any theoretical predictions less straightforward. If workers just above and below the threshold are close substitutes, profit-maximising employers should not want to pay higher total labour costs for one group over the other. However, with the statutory cost falling for both sides, each party could potentially benefit from the reduction. Employers may capture part of the savings through lower gross earnings, while employees may see higher net earnings due to

reduced deductions.

On the one hand, the side of the market with more control over setting compensation terms may be better positioned to retain a larger share of the cost reduction, similar to the logic used in the analysis of payroll tax increases. This may be especially relevant in our context, which focuses on low-earning employees that may have low bargaining power. If this is the case, employers could benefit beyond the statutory cut they received, and posted earnings would fall. On the other hand, posted earnings might be downward inflexible for low-earners as the hourly minimum wage is binding for them, preventing to an extent decreases in posted earnings. Still, whether one side benefits more, or whether the statutory split reflects the true economic incidence, remains an empirical question.

5. Empirical Strategy

We use the cohort-based roll-out of the MCB reform to estimate its short-term effects on registration and earnings. Given current data availability and the limited time since the reform was introduced, we focus on the January 2023 first wave reduction, and see its effects six months after its implementation. The design of the reform allows us to use both a regression discontinuity design (RDD) and Difference-in-Differences (DiD) approaches. At this stage, we lay out the strategy and focus on the RDD approach, with results presented in the next section.

To estimate the impact of the reform, we compare labour market outcomes around the age-based cut-off at 35, after its introduction. Because earnings and other characteristics vary with age—older workers tend to earn more due to greater experience and tenure—we cannot simply compare all workers above and below this threshold. Instead, we rely on the idea that any differences between workers just below and just above age 35 should be minimal in the absence of the reform for identification. This allows us to apply a regression discontinuity design, where the effect of the reform is identified using the following specification:

$$Y_i = \alpha + \beta \cdot \mathbb{1}(a_i < 0) + \delta \cdot a_i + \theta \cdot a_i \cdot \mathbb{1}(a_i < 0) + X_i\gamma + \epsilon_i \tag{1}$$

where a_i is the age in months of individual or cohort i, depending on the level of the

analysis (e.g., registration or different earnings definitions), normalised so that $a_i = 0$ corresponds to exactly 35 years; $\mathbb{1}(a_i < 0)$ is a dummy for being in the treated group; X_i is a vector of control variables; and β captures the discontinuity at the threshold. The model allows for different slopes of age on either side of the cut-off through the inclusion of the interaction term $\theta \cdot a_i \cdot \mathbb{1}(a_i < 0)$. Y_i is the outcome of interest for i.

For future robustness checks, we plan to include different order order polynomials using a specification of the following form:

$$Y_i = \alpha + \beta \cdot \mathbb{1}(a_i < 0) + \sum_{k=1}^{K} \delta_k \cdot a_i^k + \sum_{k=1}^{K} \theta_k \cdot a_i^k \cdot \mathbb{1}(a_i < 0) + X_i\gamma + \epsilon_i \tag{2}$$

where polynomials in a will be included to control in a flexible way for the effect of age at the time of the reform a. We will to restrict the sample to different windows around the cutt-off age. Next steps include identification checks for the RDD and implementing the DiD estimations.

6. Results

At this stage of our analysis, our preliminary results focus on the short term effects of the reform on registration, and what were its implications on different earnings definitions.

6.1. Registration

Following our conceptual analysis, the decrease in registration costs for low-earners should have increased the incentives to register. Figure 4 shows the total number of registered employees by age in months at the time of the change, for those with posted earnings below the non-reduced MCB^P in June 2023. Although registration steadily declines with age in this time window, since younger individuals are more likely to register, we do not observe any clear discontinuity increase among the younger group treated by the reform. In fact, a small discontinuity down is what seems to emerge.

[Figure 4 here]

Looking at new registrations, Figure 5 plots the accumulated number of newly registered employees entering the social security system between January and June 2023. If the reform had significantly increased registration, we would expect a visible break around age 35, but no such discontinuity appears. RDD regression analysis for new entries is still work-in-progress.

[Figure 5 here]

Overall, both the figures suggest no increase in registration among the younger, treated group, contrasting with our theoretical predictions of an increase in registration. Several reasons could be behind these results. As we are focusing on short-term effects, it is possible that changes at the extensive margin of registration take longer to materialise. Moreover, firms may have chosen to delay any administrative adjustments to their payroll systems, anticipating that the reform would soon be extended to older employees, and preferring to register all workers at once. Finally, despite the reduction in contributions, the remaining costs associated with registering new employees may still be too high to generate a noticeable change in registration behaviour.

6.2. Effects on earnings

To analyse how the reduction in registration costs was split between employers and employees, we look at how different definitions of earnings (net, posted and gross) responded after the reform. We focus on employees with posted earnings below the non-reduced MCB^P. Since our administrative data only includes posted earnings, we compute net and gross earnings based on the contribution rules introduced by the reform.

[Figure 6 here]

Panel A in Figure 6 shows an increase in net earnings increased for employees just below the age 35 threshold at the time of the change. This indicates these workers got a higher take-home pay following the lower contribution rates. The increase is confirmed in Table 4. Column 1 shows that treated workers earned about 8,427 colones more in net terms. In column 4, which uses a smaller age window, the increase is still significant

but smaller, at around 7,340 colones.

[Table 4 here]

Panel B shows the reported posted earnings, which do not display any clear change around the threshold. The line is smooth, and the regressions in columns 2 and 5 confirm that there was no significant change. This suggests no adjustment in the posted earnings declared to the social security system for workers affected by the reform.

Panel C looks at gross earnings. We see a drop at the threshold for treated employees, which means that total labour costs for employers went down. Table 4 reflects this, with column 3 showing a decrease of about 15,269 colones. In the smaller window in column 6, the drop is still large—over 10,000 colones.

Taken together, these results show that six months after the first MCB reduction for workers under 35, posted earnings remained unchanged, but we observe higher net earnings and lower gross earnings. This suggests that both employers and employees benefited from the part of the contribution that was reduced for them. In other words, the economic incidence seems to follow the statutory split, with no extra gains going to either side. While this pattern diverges from standard theoretical predictions, it is consistent with empirical evidence that highlights the importance of statutory incidence in shaping economic incidence (Saez et al. 2012; Hargaden and Roantree 2019).

It is important to note that we construct net and gross earnings based on the rules of the reform, since the administrative data only includes posted earnings and does not contain information on actual contributions. We assume that all workers earning below the MCB thresholds received the reduction, but this cannot be confirmed without access to contribution records.[10] As a result, our findings may partly reflect mechanical effects from our construction of net and gross earnings.

[10]We plan to contact the Social Security Administration to better understand take-up and explore the possibility of accessing contribution data.

7. Conclusion

This paper provides new evidence on the role of SSCs in determining registration and declared earnings for low-earning employees in Costa Rica. Motivated by low registration rates and the potentially distortionary effects of the MCB, we evaluate the short-term impacts of a 2023 reform that substantially lowered the minimum contribution floor for younger employees.

Exploiting administrative data and the age-based rollout of the reform, we find no significant effects on registration and on reported posted earnings in the first six months after the reform. However, using constructed net and gross earnings based on the reform rules, we do observe higher take-home pay for employees and lower total labour costs for employers. This suggests that the economic incidence of the reform closely followed the statutory split, with each side benefitting from the portion of the contribution reduction that applied to them. The results imply that while reducing SSC costs can ease the financial burden of registration for low earners, it may not be enough on its own to increase registration. Moreover, more time might need to pass to see the effects of the reform materialise. For policy, this suggests the potential need for complementary measures to strengthen the perceived value of registration, simplify compliance, or improve enforcement to effectively increase registration.

Moving forward, next steps include exploring additional margins of response from both workers and employers following the reform and clearly define which margins we will focus on. We will refine and complete our conceptual framework, improve our identification and empirical strategy by deepening our understanding of the system and its reform (informed by upcoming discussions with the social security administration), polish our RDD analysis including conducting robustness and identification checks, and extend our study using DiD estimations with both administrative and survey data once we gain access to panel labour force surveys.

Tables

Table 1. Effective Average Social Security Contribution Rates (2022)
(% of employees' monthly posted earnings)

	Employer	Employee
Healthcare	$\max\left(\frac{\text{MCB}^H \cdot 9.25}{\text{posted earnings}}, 9.25\right)$	$\max\left(\frac{\text{MCB}^H \cdot 5.50}{\text{posted earnings}}, 5.50\right)$
Pensions	$\max\left(\frac{\text{MCB}^P \cdot 5.25}{\text{posted earnings}}, 5.25\right)$	$\max\left(\frac{\text{MCB}^P \cdot 4.00}{\text{posted earnings}}, 4.00\right)$

Source: CCSS.

Notes: This table shows the effective average contribution rates for healthcare and pensions, by employer and employee. MCB^H is the MCB for healthcare contributions. MCB^P is the MCB for pensions contributions. Reflects how the MCB rule increases the rate for low earners below the MCB levels. The effective rate is higher for workers earning below the MCB because contributions are calculated as if they earned at least the MCB.

Table 2. Statutory Payroll Tax Rates (2022)
(% of employees' monthly posted earnings)

	Employer	Employee
Social security contributions		
Healthcare	9.25	5.50
Pensions	5.25	4.00
Mandatory private pensions	4.50	1.00
Social programmes + other institutions	7.25	
Total Rate	26.25	10.50

Source: CCSS and *Fondo de Desarrollo Social y Asignaciones Familiares* (FOSESAF).

Notes: This table shows the statutory payroll tax rates (including social security contributions) by component and by who pays them (employer or employee). The MCB rule only affects the social security contributions component. For simplicity, we only show the statutory tax rates, not the effective ones that reflect the effect of the MCB rule.

Table 3. Summary Statistics (June 2023)

	(1)	(2)	(3)	(4)
	Age < 35 Any y	Age $\geqslant$ 35 Any y	Age < 35 y < MCB^P	Age $\geqslant$ 35 y < MCB^P
Age (years)	29.96	39.71	29.69	39.70
Male (proportion)	0.59	0.59	0.61	0.59
Posted Earnings (colones)	741,877	958,582	184,475	189,263
Occupation (proportion)				
Managers	0.03	0.04	0.00	0.01
Professionals	0.15	0.15	0.05	0.05
Technicians	0.20	0.17	0.08	0.06
Clerical	0.17	0.11	0.13	0.10
Services & Sales	0.09	0.08	0.13	0.12
Agriculture & Fishing	0.02	0.02	0.01	0.01
Crafts & Traders	0.05	0.05	0.09	0.09
Operators & Assemblers	0.06	0.07	0.06	0.07
Elementary Occupations	0.19	0.18	0.44	0.45
Unclassified	0.05	0.11	0.03	0.04
Observations	360,547	301,885	39,424	29,258

Notes: This table shows summary statistics for registered workers aged 25–45 in June 2023 with only one registered job. The sample is split by age (less than 35 years and treated by the first wage of the reform, and 35 years or more and not treated by the first wave of the reform), and by whether we look at registered employees with posted earnings (y) at any level or below the MCB^P. Posted earnings are earnings before employee contributions are deducted.

Table 4. Effect of Treatment on Earnings

	(1)	(2)	(3)	(4)	(5)	(6)
	Age 25-45 Net	Age 25-45 Posted	Age 25-45 Gross	Age 31–38 Net	Age 31–38 Posted	Age 31–38 Gross
treat	8427.0***	-1008.6	-15268.7***	7340.5***	392.0	-10109.9***
	(1387.3)	(1425.9)	(1487.2)	(2398.5)	(2444.3)	(2520.8)
Observations	68,682	68,682	68,682	23,947	23,947	23,947

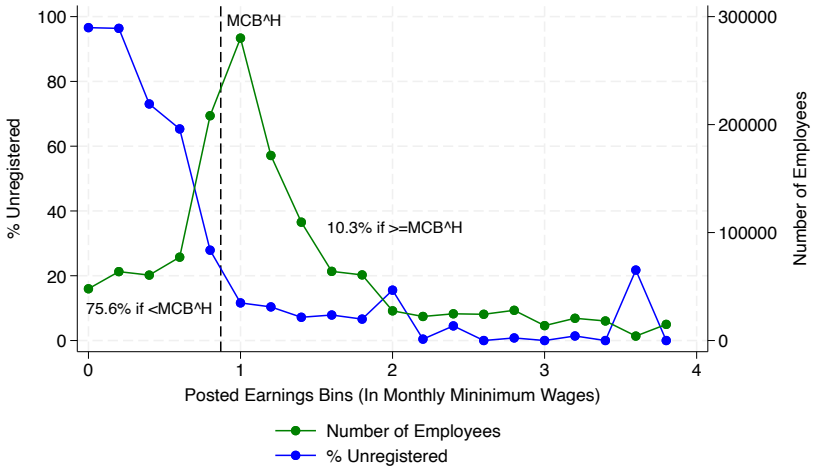
Notes: This table shows regression discontinuity estimates of the MCB reform's effect on different earnings definitions. We report outcomes for net earnings (after employee contributions), posted earnings (before deductions), and gross earnings (total labour costs). The sample includes workers with posted earnings below MCB^P. Columns 1 to 3 use the 25-25 age bandwidth, while Columns 4-6 the 31-38 age bandwidth. Robust standard errors in parentheses. $p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5. Descriptive Statistics by Sample

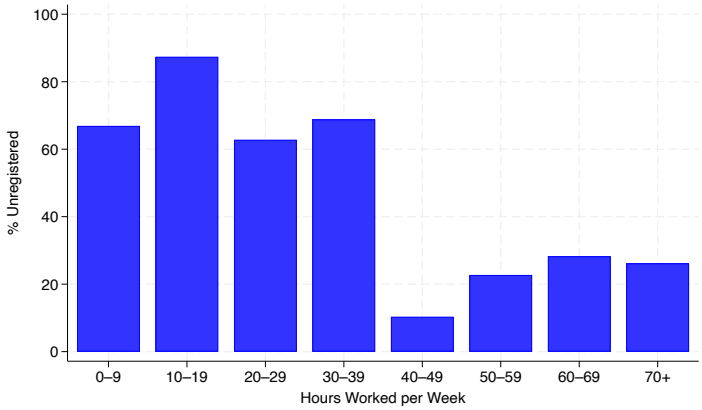
	(1)	(2)	(3)	(4)	(5)
	Sample 1 All ages Any y	Sample 2 Age < 35 Any y	Sample 3 Age $\geqslant$ 35 Any y	Sample 4 Age < 35 < MCB^P & P	Sample 5 Age $\geqslant$ 35 < MCB^P & I
age_years_em	34.61	30.04	39.76	29.92	39.92
male	0.63	0.61	0.64	0.55	0.52
salario	901100.41	785032.69	1031826.78	161043.19	160213.52
Fuerzas armadas	0.04	0.03	0.05	0.00	0.01
Directores y gerentes	0.04	0.03	0.06	0.00	0.01
Profesionales	0.16	0.16	0.17	0.09	0.14
Técnicos y prof	0.21	0.22	0.20	0.10	0.08
Personal de apoyo	0.16	0.18	0.13	0.19	0.15
Trabajadores servicios y vendedores	0.09	0.09	0.08	0.16	0.14
Agricultores	0.00	0.00	0.00	0.00	0.00
Oficiales, operarios	0.05	0.05	0.06	0.06	0.06
Operadores de máquinas	0.06	0.05	0.07	0.03	0.05
Ocupaciones elementales	0.18	0.18	0.18	0.35	0.36
fraq_lbmc	0.11	0.12	0.11	1.00	1.00
partial	0.06	0.06	0.06	1.00	1.00
partial_bbmc	0.04	0.03	0.04	1.00	1.00
Observations	4873200	2583360	2289840	86418	84364

Figures

Figure 1. Unregistered Employees (IV Quarter 2022)



A. By Posted Earnings

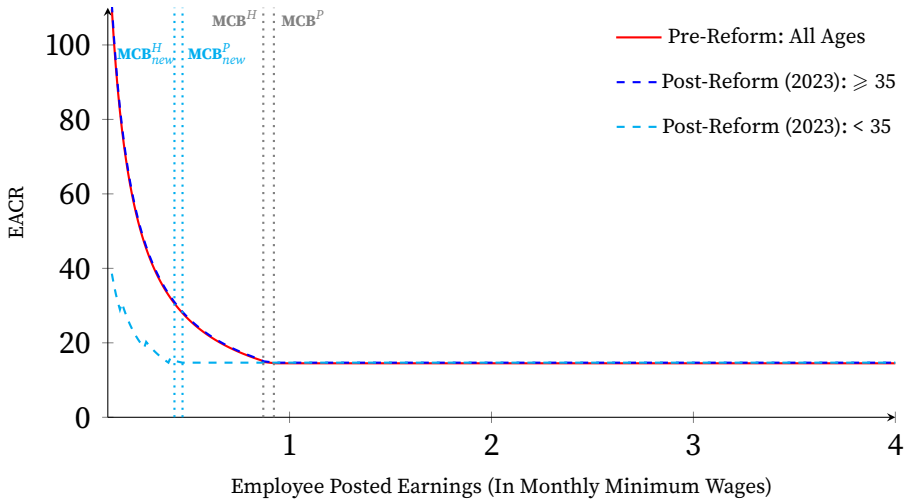


B. By Hours Worked per Week

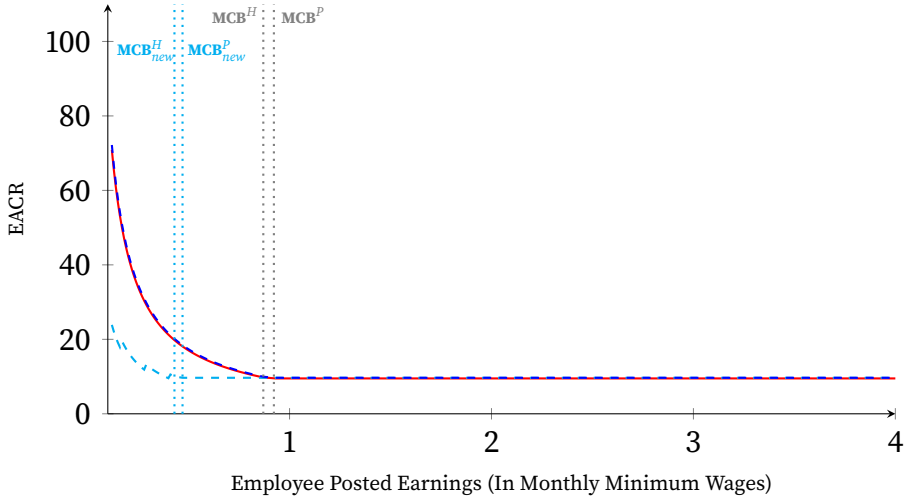
Source: *Encuesta Continua de Empleo* (ENAHO), IV Quarter 2022

Notes: This figure shows the share of employees not registered with social security, based on survey data. Panel A shows unregistered employee rates, as well as the total number of employees, by bins earnings. Panel B shows unregistered rates by hours worked.

Figure 2. Effective Average Contributions Rates, Pre- and Post-Reform



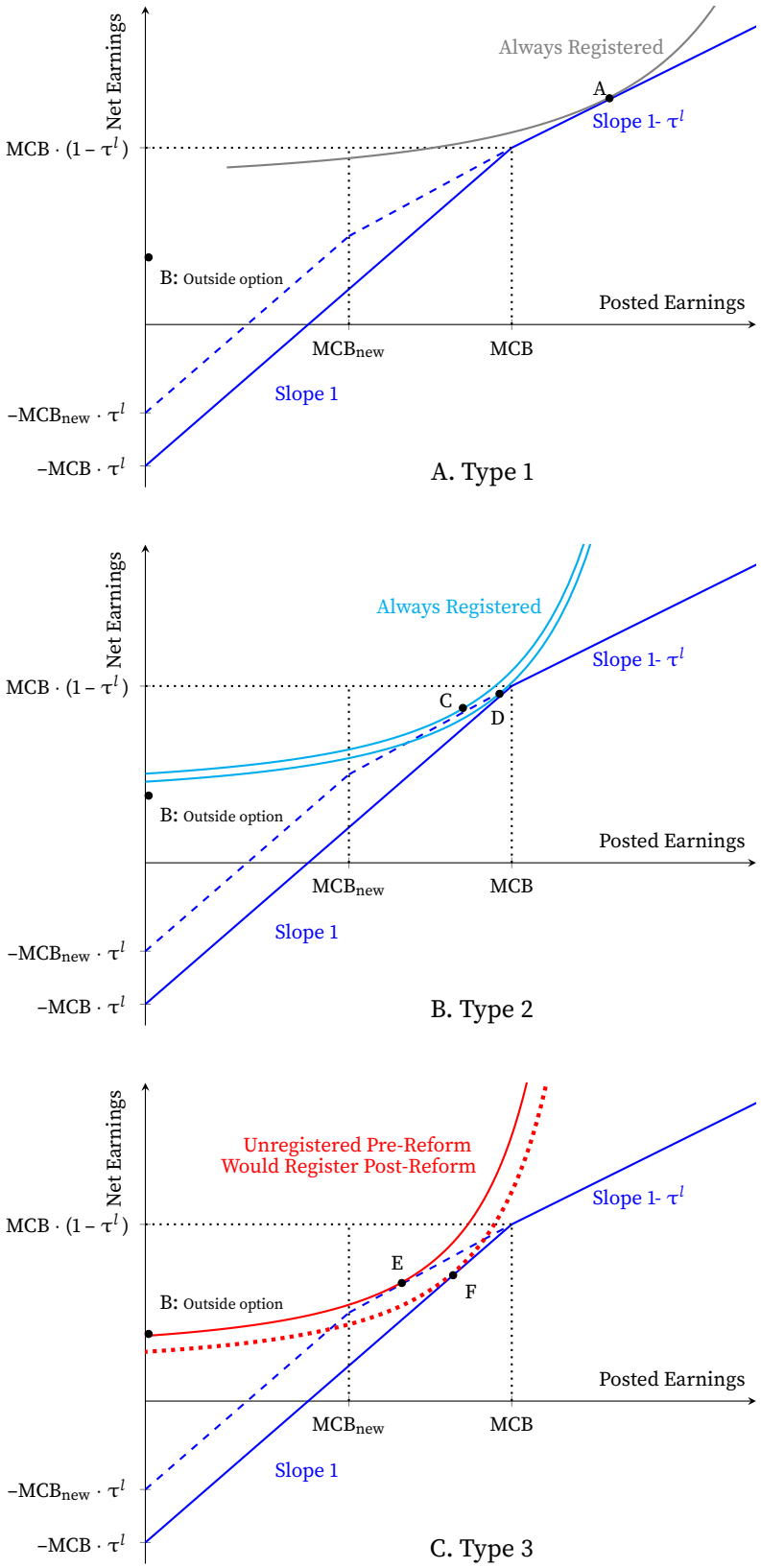
A. For Employers



B. For Employees

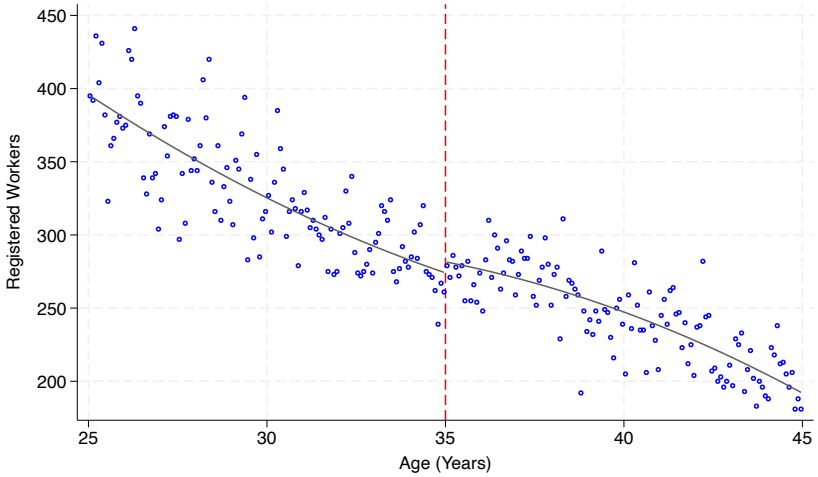
Notes: This figure shows how effective average contribution rates change across employee posted earnings before and after the reform that lowered the MCBs. Panel A shows employer rates. Panel B shows employee rates. Solid red lines are pre-reform; dashed blue lines are post-reform (for treated and not treated age groups). MCB^H is the MCB for healthcare contributions, and MCB^H_{new} the MCB for healthcare contributions for treated groups after the reform. MCB^P is the MCB for healthcare contributions, and MCB^P_{new} the MCB for healthcare contributions for treated groups after the reform. Irregularities in the dashed blue line below the new MCB reflect the new more-proportional design for pension contributions.

Figure 3. Budget Set Diagrams for Labour Supply Decisions



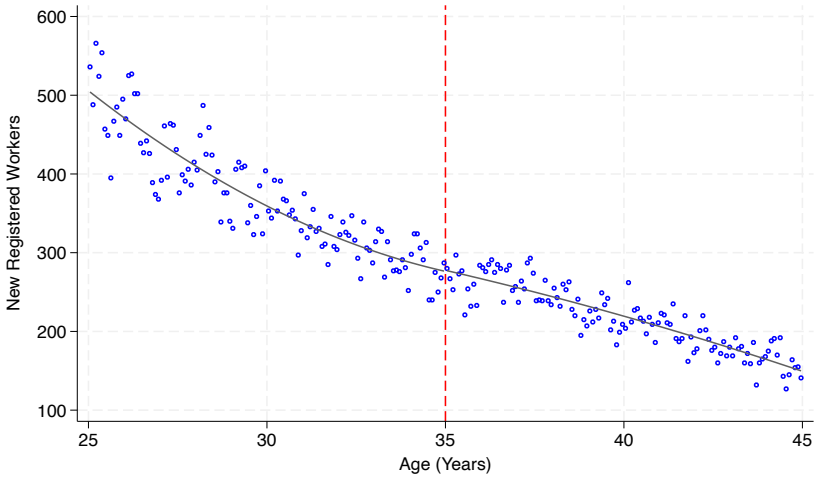
Notes: This figure shows theoretical budget sets and labour supply choices for three types of workers, before and after the MCB cut. Dark blue lines show pre- and post-reform budget sets, with kinks at MCB and MCB$_{new}$. Curves show indifference curves based on utility (Type 1:gray, Type 2: light blue, Type 3:red). Posted earnings are on the x-axis, net earnings (take-home pay) on the y-axis.

Figure 4. Registration with Social Security System



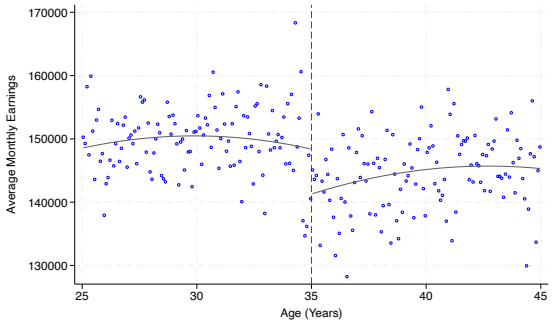
Notes:This figure shows the number of registered employees in June 2023, by age bins. It includes only employees with posted earnings below MCB^P. The vertical line at age 35 marks the reform first wave cut-off.

Figure 5. New Registrations with the Social Security System

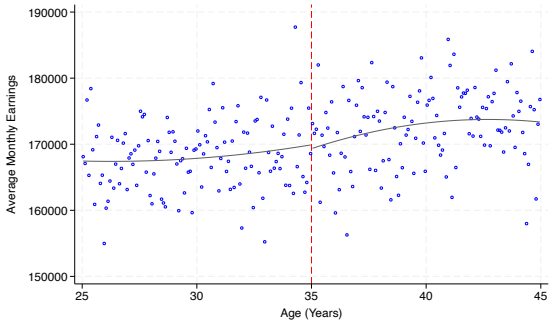


Notes: This figure shows the number of new registrations between January and June 2023, by age bins.It includes only employees with posted earnings below MCB^P.The vertical line at age 35 marks the reform first wave cut-off.

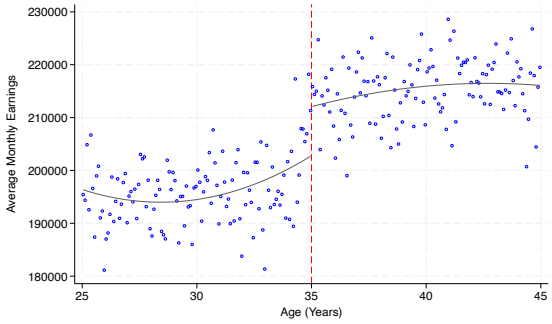
Figure 6. Earnings



A. Net Earnings



B. Posted Earnings



C. Gross Earnings

Notes: This figure shows earnings by age bins in June 2023. Panel A shows net earnings (after employee SSC), Panel B shows posted earnings (before employee SSC), and Panel C shows gross earnings (posted earnings + employer SSC). All are for workers with posted earnings below MCB^P. Since administrative data only include posted earnings, net and gross earnings are computed by the authors using statutory contribution formulas, based on the reform's rules.

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